Your PHY4605 Capstone Investigation

Real problems. Real tools. Real physics.

Scenario

A real-world problem to investigate.

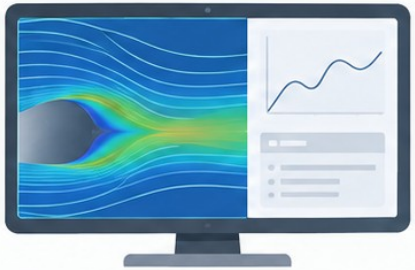


- One of ten distinct problem briefs
- Grounded in real systems and real constraints
- Work in a team to define and explore the problem

model

Simulation

AI-assisted realistic analysis.



- Use AI-assisted tools to build a realistic simulation
- Include the key physics that matter
- Make and test reasonable choices and assumptions

evidence

Evidence

Show, compare and justify your results.



- Compare with experimental or real-world data
- Validate that your model reproduces key observed behaviour
- Build your 15% course component through staged evidence

Ten groups, ten distinct problem briefs
Different systems. Shared goal: deeper understanding through physics.

A 15% course component
Built through staged evidence.

The New Capstone Contract

Ten different problem briefs. Many possible approaches. Your investigation, your responsibility.



Start from a detailed physical scenario
and broad objective



- Each group receives a different real-world brief.
- Interpret the **physical scenario** and define the problem in your own words.
- Set a sensible **model scope** to meet the objective.

There is no common starter Live Script



- You are not given a starter Live Script.
- There is **no prescribed solution path.**
- You choose the physics, numerical method, and algorithm that fit your brief.

Your group makes and defends the modelling decisions



- State and justify your **assumptions** and model scope.
- Generate and interpret **evidence.**
- Discuss **validation, limitations** and **reproducibility.**
- Keep an **AI decision record** of how AI tools were used.

A defensible investigation is the goal. Be physically faithful, relevant and transparent about what is included and what is omitted.

What the Brief Gives You — and What You Must Decide

The brief defines the problem boundary. You decide how to model, investigate and justify it.

What the Brief Gives You

A real-world problem with an open interpretation.



Context
Real-world situation and motivation.

Process
What is happening in the physical system.

Initial information or data
Available details, measurements or references.

Broad objective
What you are expected to achieve.

Your investigation

From brief to defensible solution

What You Must Decide

Your group makes the modelling and investigation choices.

Model scope
What to include and what to omit.

Physical effects
Which additional physics are relevant.

Assumptions
State and justify your assumptions.

Numerical method and algorithm
Choose and design a method and implement the code.

Outputs and evidence
What results to produce and how to present evidence.

Validation and limitation
How to check the model and what its limits are.

Reproducibility
Organise and document a repeatable investigation.

AI decision record
Record and justify how AI was used.

Ask questions when the boundary is ambiguous

If anything in the brief is unclear, discuss it with your lecturer before making key decisions.

Realism Is the Goal, Not Complexity

The strongest model is the most defensible approximation to the real process.

Simple Lecture-Level Model

Capture the core physics.



- Idealized system
- Few assumptions
- Analytic understanding
- Useful baseline predictions

baseline

Relevant Physical Effects

Add what matters for this problem.

Dissipative effects
e.g. drag, losses

Environmental influence
e.g. medium, boundaries

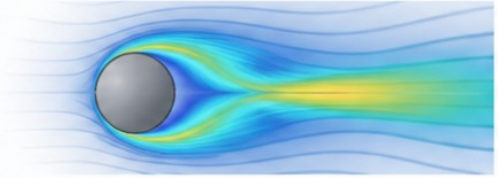
System details
e.g. geometry, material response

Coupled processes
e.g. thermal, electrical, chemical

evidence

Defensible Realistic Simulation

Justify choices and check against reality.



Evidence
Compare with experimental or real-world data

Check
Confirm the model reproduces key observed behavior

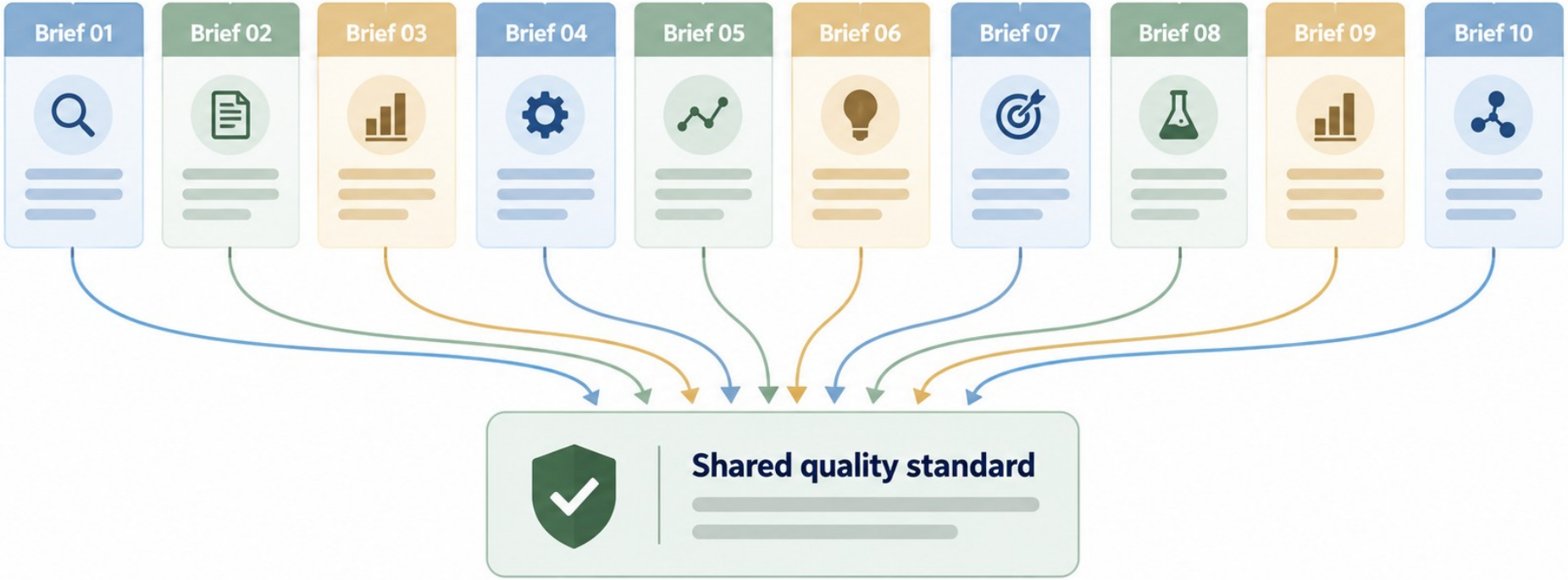
Defend
Explain included effects and omitted effects

More terms ≠ better model

Include the effects that materially change the result, and justify why the rest can be omitted.

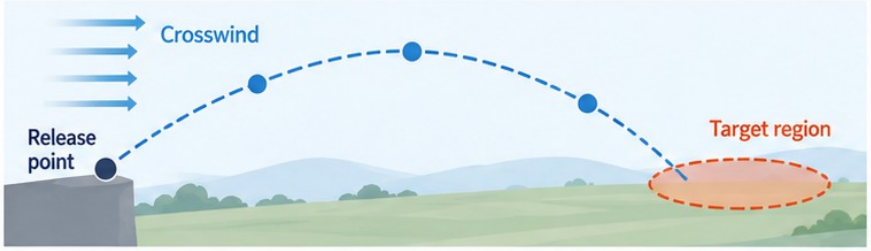
One Brief per Group, Ten Distinct Investigations

- Each group receives **one assigned brief**
- Scenarios, data and objectives **are distinct**
- The same high-level standard **applies to every group**
- **Do not copy** a common solution route



Projectile Investigations: Wind, Drag and Terrain

BRIEF 01 — Projectile Delivery in Crosswind with Quadratic Drag



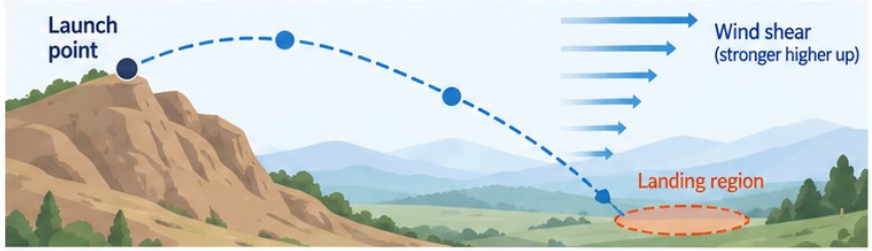
Context + objective: A package or test body moves through crosswind toward an intended delivery region; investigate how convincingly the delivery outcome can be represented.

Possible realism dimensions: 3D motion, relative airflow, varying crosswind, aerodynamic properties, release uncertainty, air-density variation, lift or spin.

Likely information categories: Launch or release conditions, object properties, target region, wind measurements or ranges, and any reference-trial observations actually supplied.

Boundary questions: Is a steady crosswind adequate? Is 3D motion material? Can the body be treated as a point? Which aerodynamic effects are supported by evidence?

BRIEF 02 — Projectile Launch from Elevated Terrain under Wind Shear



Context + objective: An object launches from elevated or uneven terrain while wind changes with height; investigate whether it can reach an intended or acceptable landing region.

Possible realism dimensions: Wind shear, terrain profile, non-level impact surface, launch variability, atmospheric variation and local obstructions.

Likely information categories: Site elevations or terrain information, wind observations at selected heights, launch conditions, object properties, and an intended landing or safety region.

Boundary questions: How much terrain is needed? Is 2D defensible? What happens outside measured wind heights? Are gusts, turbulence or obstacles material but unsupported?

Aim for defensible realism: preserve a transparent baseline; add effects that materially affect the objective and can be supported or tested; justify important omissions.

Circuit Investigations: Non-Ideal Charging and Loading

Explore real-world circuit behaviour, choose a defensible model scope, and justify your assumptions.

BRIEF 03 — RC Charging with Leakage and Temperature-Dependent Components

Context + objective

A capacitor circuit operates where leakage and temperature may alter charging; investigate whether its intended timing, storage or threshold function remains credible.

Possible realism dimensions

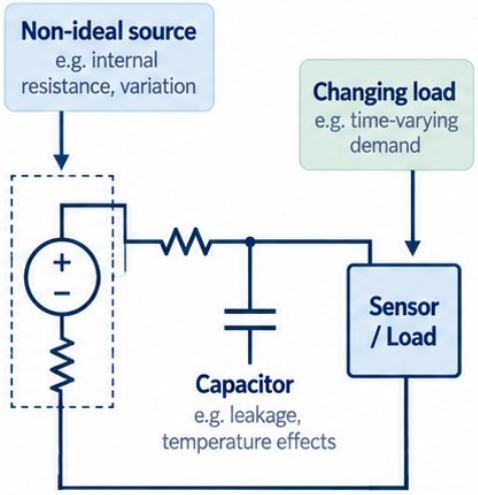
Leakage pathways, temperature-dependent component behaviour, tolerance, supply variation, measurement loading, ageing or self-heating.

Likely information categories

Nominal component specifications, tolerance or temperature ranges, supply information, temperature records, observed charging data if supplied, and the usable operating range.

Boundary questions

Is temperature imposed or coupled to circuit heating? Which variations matter? Are parasitics supported? Over what range are the stated properties credible?



BRIEF 04 — RC Sensor Circuit with Internal Source Resistance and Time-Varying Load

Context + objective

A sensor is supplied by a non-ideal source while its connected load changes; investigate whether the signal remains usable and physically credible.

Possible realism dimensions

Internal source resistance, changing load demand, sensor output behaviour, wiring and measurement loading, switching transients and sensor noise.

Likely information categories

Source, sensor and load specifications; a load schedule or trace if supplied; calibration readings, sampling information, and an acceptable signal range.

Boundary questions

Are load changes abrupt or gradual? Does source regulation matter? Which parasitics are material? Is noise central or secondary to the objective?

Aim for defensible realism

Preserve a transparent baseline, add effects that materially affect the objective and can be supported or tested, and justify important omissions.

Radiation Investigations: Detector Effects and Mixed Sources

BRIEF 05

Radioactive Decay with Background Radiation and Detector Dead Time

Context + objective

A detector records a decaying source plus unrelated background events while closely spaced events may be missed; investigate how reliably source behaviour can be inferred.

Possible realism dimensions

Background variation, detector dead time, efficiency, finite counting intervals, source–detector geometry, shielding and random counting variation.

Likely information

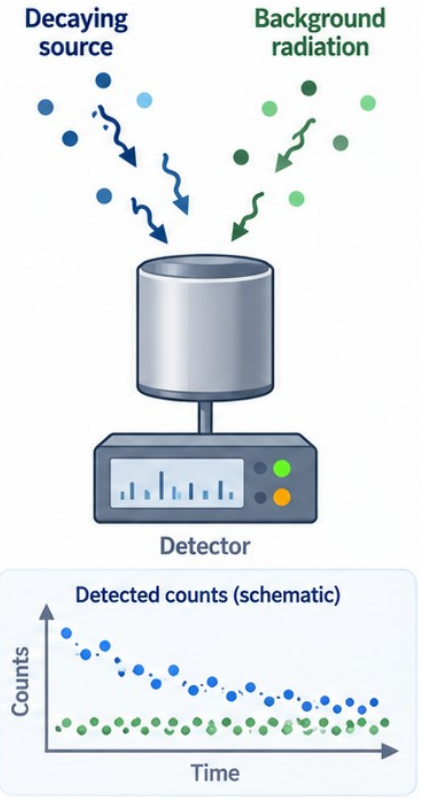
Time-stamped or interval counts, background-only observations, detector specifications or calibration information if supplied, source references and measurement geometry.

Boundary questions

Is background stable? Is the dead-time description adequate? Can efficiency and geometry be treated as constant? Are observations near a detector-response limit?

Scope cue

Select and justify the effects that matter for the objective.



BRIEF 06

Two-Isotope Decay Mixture with Efficiency and Counting Noise

Context + objective

Measurements contain two radioactive components with different timescales; investigate what can be defensibly be learned about their combined behaviour or composition.

Possible realism dimensions

Different decay timescales, isotope-dependent efficiency, background counts, finite intervals, repeated observations and counting variability.

Likely information

Count records, measurement times, candidate isotope information, efficiency or calibration evidence if supplied, background observations and repeated measurements if available.

Boundary questions

Are isotope identities known? Can the detector distinguish signals? Is efficiency stable? Is dead time material? How much uncertainty can the data support?

Scope cue

Do not accumulate features; defend the chosen scope.

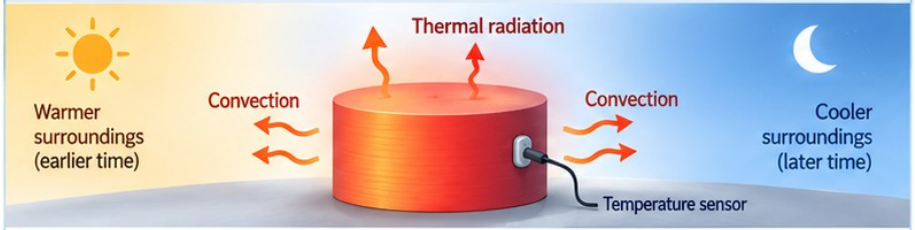
Aim for defensible realism: preserve a transparent baseline; add effects that materially affect the objective and can be supported or tested; justify important omissions.

Thermal Investigations: Changing Surroundings and Internal Heating

Two capstone briefs for Year-2 physics students

BRIEF 07 — Cooling with Thermal Radiation and a Changing Ambient Temperature

A hot object cools in changing surroundings where radiation may matter; investigate its thermal history relative to a useful, safe or observable temperature range.



Context + objective

A hot object cools in changing surroundings where radiation may matter; investigate its thermal history relative to a useful, safe or observable temperature range.

Possible realism dimensions

Changing ambient conditions, convection, radiation, surface properties, geometry, environmental exposure and sensor response.

Likely information

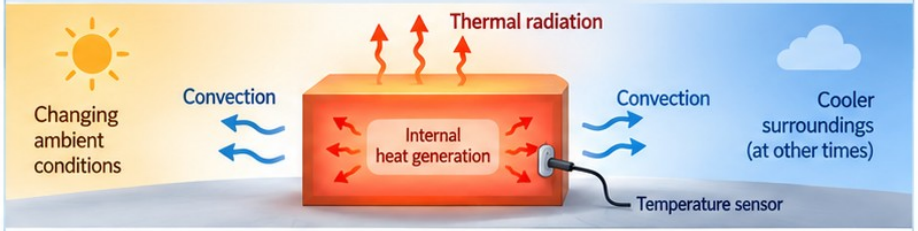
Initial and observed temperatures, ambient-temperature records, object dimensions, mass and material information, surface-property estimates, and any reference cooling run supplied.

Boundary questions

Is a uniform object temperature defensible? Do radiative surroundings match measured air temperature? Which heat-loss pathways matter? Are gradients, moisture or phase changes supportable?

BRIEF 08 — Cooling of a Heated Object with Internal Heat Generation

An object continues producing heat while losing energy; investigate whether its operating temperature remains acceptable and how behaviour changes during operation.



Context + objective

An object continues producing heat while losing energy; investigate whether its operating temperature remains acceptable and how behaviour changes during operation.

Possible realism dimensions

Time-dependent or temperature-dependent heating, convection, radiation, conduction through supports, insulation, ambient variation and control action.

Likely information

Power or operating records, measured temperatures, ambient conditions, geometry and thermal properties, operating limits and reference observations if supplied.

Boundary questions

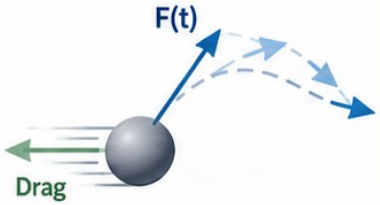
Is heating spatially uniform? Can one temperature represent the object? Does temperature affect generation? Which support or enclosure pathways matter? Is control relevant?

Aim for defensible realism: preserve a transparent baseline; add effects that materially affect the objective and can be supported or tested; justify important omissions.

Force Investigations: Direction, Drag, Mass and Measurement

Two real-world scenarios. Different physics, different choices, no prescribed solution path.

BRIEF 09 — Impulse on a Moving Body with Drag and Changing Force Direction



A moving body experiences a finite-duration force whose direction changes while resistance acts. Investigate how the intervention alters subsequent motion.

Context + objective

A moving body experiences a finite-duration force whose direction changes while resistance acts; investigate how the intervention alters subsequent motion.

Possible realism dimensions

Force magnitude and direction history, drag, initial motion, dimensionality, body orientation, rotation and force-measurement uncertainty.

Likely information categories

Force and direction records or ranges, event duration, body properties, initial motion, environmental information and any motion-tracking observations supplied.

Boundary questions

Is translation sufficient? Does the force act through the centre of mass? Is mass constant? Is this a short impulse or sustained actuation? Is the drag representation credible?

BRIEF 10 — Time-Varying Directional Actuator Force with Mass Change and Sensor Uncertainty



An actuator changes force direction while system mass changes and sensors are imperfect. Investigate how confidently motion or actuator performance can be predicted.

Context + objective

An actuator changes force direction while system mass changes and sensors are imperfect; investigate how confidently motion or actuator performance can be predicted.

Possible realism dimensions

Commanded versus realised force, physical mass-change process, sensor bias/noise/delay, actuator limits, misalignment and external resistance.

Likely information categories

Actuator commands or force records, orientation information, mass or payload history, sensor specifications and calibration evidence if supplied, initial state and reference tracking data.

Boundary questions

What enters or leaves as mass changes? Does transfer carry momentum? Are rotational effects material? Which sensor imperfections affect the objective? Is 2D defensible?

Aim for defensible realism | Preserve a transparent baseline; add effects that materially affect the objective and can be supported or tested; justify important omissions.

From a Lecture Model to a Real Process

Add physics in a disciplined way — not everything, and not arbitrarily.

1. Lecture-Level Model

A familiar starting point.



- Idealised process
- Few assumptions
- Analytic understanding
- Useful baseline prediction

Identify a missing effect

2. Is the extra physics relevant?

Consider three questions before you add it.

Relevance
Does it materially affect the result for this scenario?

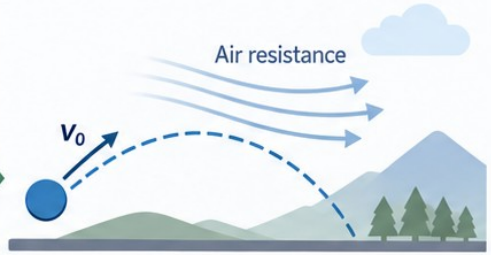
Data
Is there enough information to support it?

Computability
Can you implement and test it with available tools?

Choose a defensible model level

3. Richer Process Model

A closer approximation to the real world.



- Additional physical effects
- More realistic representation
- Test and validate against evidence
- State assumptions and limitations

Start from a taught process and identify a missing effect

Judge its relevance, available data and computability

Choose a model level you can test and defend

Keep the baseline visible for comparison

Your Investigation Reasoning Chain

From a real-world scenario to a reproducible handoff.

1. Understand and Define
Interpret the problem and set the scope.

1 Scenario
Read and interpret the physical scenario.

2 Model
Choose model scope and state key assumptions.

2. Build and Implement
Turn your model into a computation.

3 Algorithm
Select a numerical method and design your algorithm.

4 Implementation
Write and run your code. Document choices including AI use.

3. Test and Analyse
Generate evidence and draw conclusions.

5 Tests and Evidence
Run tests and compare with relevant evidence.

6 Interpretation
Explain what your results mean in the real-world context.

4. Report and Handoff
Be explicit about limits and ensure reproducibility.

7 Limitations
State what is not included and what could change the result.

8 Reproducible Handoff
Provide a clear, reproducible investigation with an AI decision record.

Validation loop
Use evidence to refine your model and assumptions.

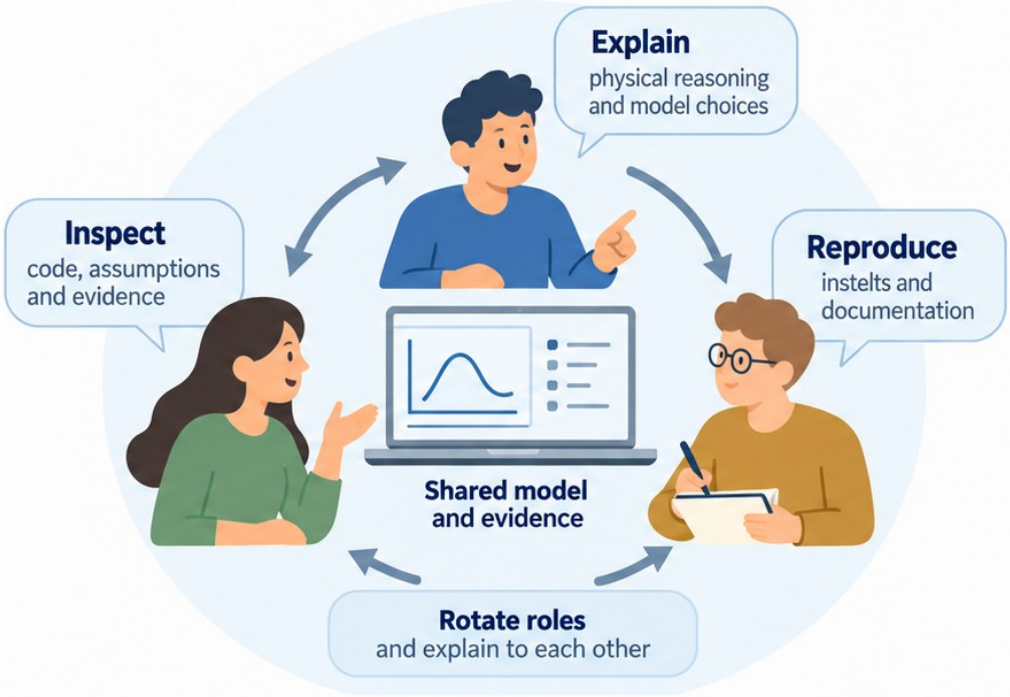
New insights may lead you to revisit earlier choices.

Key points

1. Scenario to physical interpretation
2. Model choices to algorithm and implementation
3. Tests and evidence to interpretation and limitations
4. Finish with a reproducible handoff

How Groups Should Work

Share the work, build understanding, and produce a defensible **investigation.**



Share physical reasoning, implementation, evidence and documentation

Rotate explanations instead of dividing the work into silos

Every member understands the final model and code

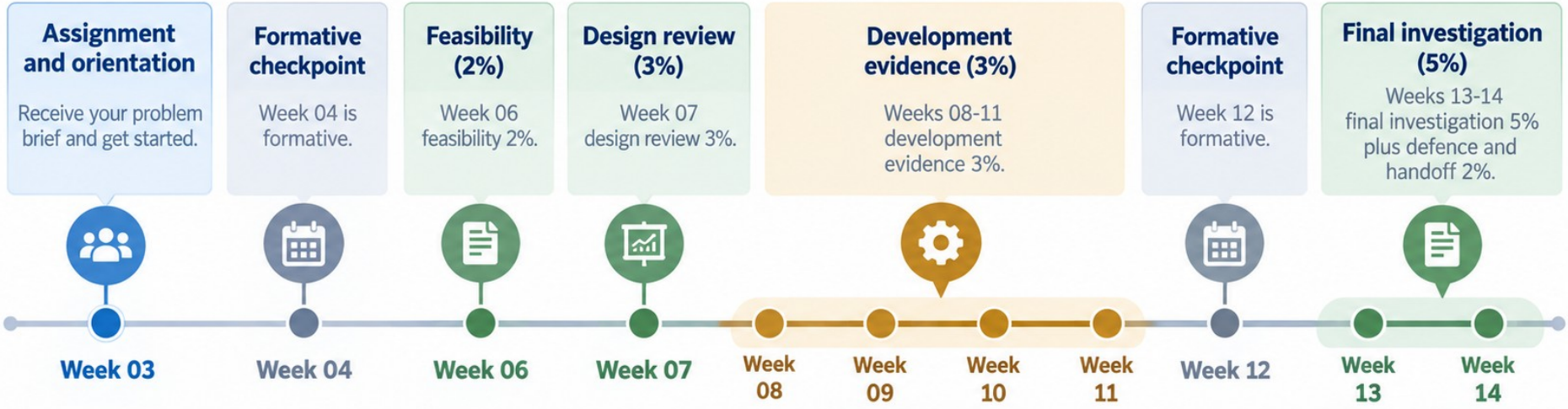
Record concise AI decisions and independent checks

A shared investigation, not individual silos

Collaborate, question, and teach each other — so everyone can defend the final work.

Capstone Milestones and Weighting

A large investigation becomes manageable when you deliver staged evidence.



Assessed milestone
Contributes to 15%

Formative checkpoint
Not assessed

Development period
Assessed as a whole (3%)

Total capstone weighting
15%

Focus on steady progress and defensible evidence

Use the milestones to plan, document and demonstrate your best investigation.

Week 04: Understand the Assigned Brief

Turn the assigned title into a precise investigation question before implementation.

1 Restate the scenario and broad objective in your own words

- What is the physical scenario about?
- What is the broad objective?
- How would you explain it simply?

2 Identify available information, data and unknowns

- What information is provided?
- What data (if any) are available?
- What key quantities are unknown or uncertain?

Assigned Brief

Real-world physics problem



Context | Constraints | Deliverables

3 Clarify the real-world prediction or decision

- What real-world prediction or decision is needed?
- Who will use the result?
- What would make the result useful and relevant?

4 Record ambiguity and questions for the lecturer

- What parts are unclear or ambiguous?
- What assumptions need discussion?
- What questions do you want to ask the lecturer?

Formative Checkpoint

A clear and shared understanding of the brief now will save time and lead to a more focused and defensible investigation later.

Week 06: Feasibility Proposal — 2%

Show that your investigation is manageable, realistic and defensible.

2%
of final mark

1. Model Direction

Propose a model direction and likely realism extension.

- State the physical scenario in your own words.
- Define the model scope (what is included and excluded).
- Identify at least one likely extension to increase realism.
- Justify why this is a sensible and manageable approach.



2. Data and Assumptions

List data, parameters and assumptions.

Data

What data will you use? (e.g. experimental, published, open sources)

Parameters

What key parameters are needed and how will you obtain them?

Assumptions

List your main assumptions and explain why they are reasonable.

3. Computational Route

Outline the computational route and verification intention.

Numerical method

What kind of numerical method(s) do you plan to use? (e.g. time-stepping, iteration, solver, etc.)

Algorithm

Outline the main steps of your algorithm at a high level.

Implementation plan

What are the key tasks and milestones before the design review?

4. Verification Intention

Outline how you will check your results.

Evidence

What evidence will show that your model is working as intended?

Validation

How do you plan to verify key aspects of the model (e.g. benchmarks, limiting cases, comparison data)?

Limitations

What are the main limitations and risks, and how might you address them?

No final code or final method is required yet.

This is a feasibility proposal, not the final investigation.
It should demonstrate a clear, realistic and manageable plan.

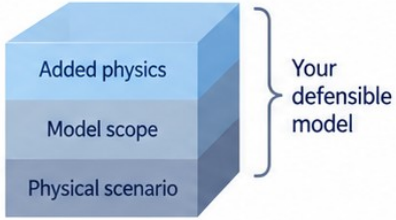
Week 07: Model and Computational Design Review — 3%

Show that your planned approach is physically faithful, feasible and defensible.

3%
of final mark

Model and Scope

Justify model scope and added physics.



- Define what is included and why
- Explain key simplifications
- Show the model is appropriate for the real process

Data and Assumptions

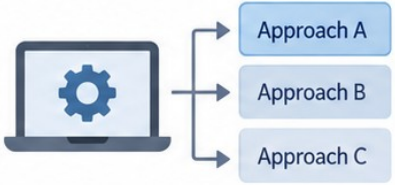
Identify data needs and assumptions.



- What data are required?
- What are the key assumptions?
- How will you obtain or estimate missing information?
- State the expected limitations due to data and assumptions

Numerical Approach

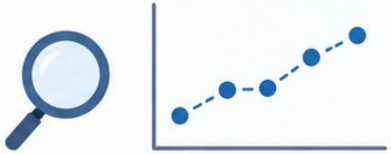
Justify a numerical approach without treating it as the only possible route.



- Explain your chosen approach
- Justify why it is suitable
- Acknowledge reasonable alternatives
- State expected computational requirements

Testing and Evidence

Plan testing, validation and sensitivity.



- How will you test the implementation?
- What evidence will show the model is behaving as expected?
- Plan validation and sensitivity studies
- Define what would challenge your conclusions

Approval means defensible and feasible, not final correctness.

- ✓ Physically reasonable and well justified
- ✓ Practical to implement within the remaining time
- ✓ Clear plan for testing, validation and sensitivity

Weeks 08–11: Build Evidence Through Iteration — 3%

Make steady progress, test your ideas, and build a defensible case.



3% Development Evidence

Show your iteration and learning through evidence, not a retrospective chat diary.

Week 12: Freeze, Reproduce and Rehearse

Make the final readiness criteria concrete before presentations begin.

Week 12
Formative

Finalisation Checklist

Be ready to hand in and present a reproducible investigation.

Freeze the final model and evidence
Decide what is included, and keep it fixed for submission and presentation.

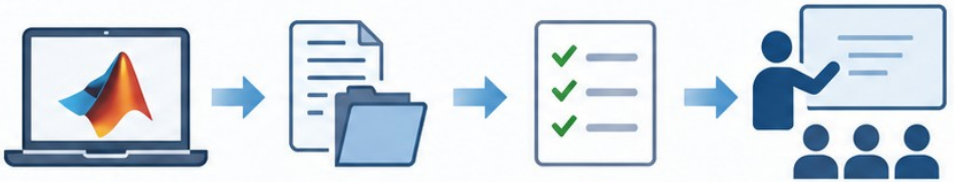
Rerun cleanly from a fresh MATLAB session
Follow your documented steps to reproduce all results from scratch.

Check units, limiting behaviour, sensitivity and plausibility
Look for mistakes and test that results make physical sense.

Prepare reproducibility and individual explanation
Organise files, instructions and an AI decision record, and rehearse your individual defence.

Rerun from a Fresh MATLAB Session

Demonstrate that your work is reproducible on a clean setup.



1. **Start fresh**
   Open a new MATLAB session with default settings.
2. **Use your submission files**
   Follow your documented steps (e.g. README) to run the model.
3. **Check key aspects**
   Units, limiting behaviour, sensitivity and plausibility.
4. **Rehearse your explanation**
   Be ready to explain your model, choices, evidence, limitations and next steps.

A clean rerun builds confidence

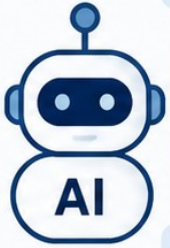
If you can reproduce it, explain it, and defend it, you are ready for the final presentation.

AI Tutorial: Use AI as a Modelling Partner, Not a Validator

AI can help you explore, plan and code, but you must independently validate the results.

Use AI for Productive Support

Get ideas, options and practical help.



- Interpret a scenario and compare model levels
- Ask for algorithm plans, code drafts, debugging and documentation help
- Use MATLAB Copilot or ChatGPT for alternatives, not authority

ideas and options

Your Modelling Work

Make and justify your own decisions.

- Define model scope and assumptions
- Design algorithm and numerical method
- Implement and test a computation
- Interpret results and state limitations

your results

Independent Validation

You are responsible for the checks.

- Validate units and dimensions
- Check limits, sensitivity and physical meaning
- Compare with references or real-world behaviour (as appropriate)
- Confirm the model is fit for purpose

AI is never the validator.
You must do these checks yourself.

Record AI Decisions

Be transparent and critical.

- **Accepted** — Useful suggestion, adopted (with notes).
- **Modified** — Used with changes (explain why).
- **Rejected** — Not used (explain why).
- **Keep an AI decision record** — Include prompts, key suggestions and your rationale.

Use AI to go further, not to replace your judgement

AI can suggest possibilities.
You decide what is physically valid.

Safe Prompts, Failure Modes and Your Next Action

Use AI constructively, stay alert to common pitfalls, and take the next step with confidence.

Safe Prompt Patterns

Ask AI to expose assumptions, units and testable alternatives.

Clarify the scenario
"List the key physics, assumptions and model scope for this problem."

Check consistency
"What are the relevant units, boundaries and conservation laws?"

Explore alternatives
"Suggest different modelling approaches and their trade-offs."

Plan how to test
"What evidence would support or refute this model?"

Common Failure Modes

Watch for invented values, unstable steps and unsupported physics.

Invented or unverifiable content
Plausible-sounding numbers, references or data that are not real.

Unstable or flawed steps
Algorithms that seem reasonable but are numerically unstable or incorrect.

Unsupported physics
Claims or simplifications that lack a sound physical basis.

A polished plot can still be unverified
A neat figure does not guarantee correctness or validation.

Your Next Action

Read, frame and question the assigned brief.

1. **Read the brief carefully**
   Understand the physical scenario, context and objectives.
2. **Frame your approach**
   Identify the key questions, model scope and what you need to find out.
3. **Question and explore**
   Challenge assumptions, consider alternatives, and note what will be hard.
4. **Prepare to discuss**
   Bring your initial ideas and questions to your group and next week's class.

Next: read, frame and question the assigned brief.

A successful capstone is a defensible, physically faithful answer to a real problem, not the most complicated code. Use AI wisely, and take the next step.